Auteur: Raj Shah
Studierichting: Informatica
Oefeningenreeks 5A nr. 9.2

$$\int_1^2 (x^2+4)^{\frac{1}{3}\cdot xdx}$$

Stel $x^2 + 4 = t \Rightarrow dt = 2xdx \Leftrightarrow dx = \frac{dt}{2x}$

$x_1 = 5$ en $x_2 = 8$

$$= \frac{1}{2} \cdot \int_5^8 t^{\frac{1}{3}} dt$$

$$= \frac{1}{2} \cdot \left[\frac{t^{\frac{4}{3}}}{\frac{4}{3}} \right]_5^8$$

$$= \frac{1}{2} \cdot \left(\left(\frac{3 \cdot 8^{\frac{4}{3}}}{4} \right) - \left(\frac{3 \cdot 5^{\frac{4}{3}}}{4} \right) \right)$$

$$= \frac{1}{2} \cdot \left(\frac{24\sqrt[3]{8}}{4} - \frac{15\sqrt[3]{5}}{4} \right)$$

$$= \frac{1}{2} \cdot \left(\frac{48}{4} - \frac{15\sqrt[3]{5}}{4} \right)$$

$$= 6 - \frac{15\sqrt[3]{5}}{8}$$

References